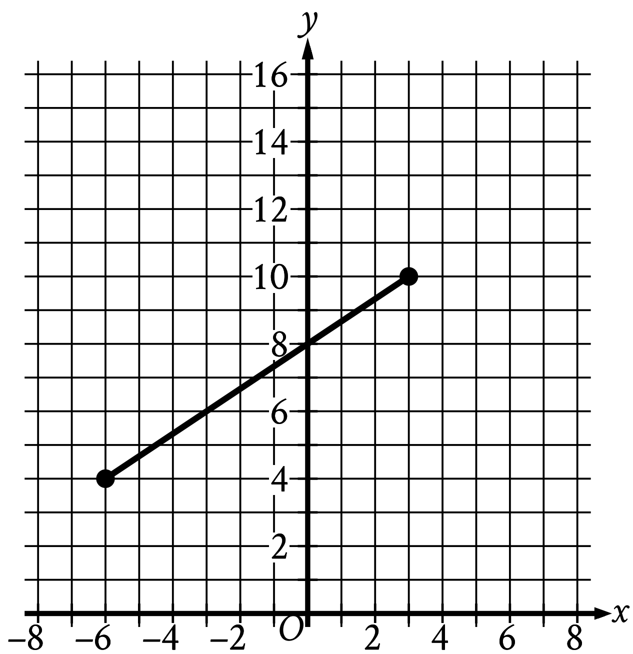
HARD · GEOMETRY AND TRIGONOMETRY

Area and volume

01

10 questions · ~15 min



- The line segment slants gradually up from left to right.
- The line segment begins at the point (negative 6 comma 4).
- The line segment ends at the point (3 comma 10).

The line segment shown in the *xy*-plane represents one of the legs of a right triangle. The area of this triangle is $36\sqrt{13}$ square units. What is the length, in units, of the other leg of this triangle?

A. 12

B. 24

C. $3\sqrt{13}$

D. $18\sqrt{13}$

Q2

GRID-IN GEOMETRY AND TRIGONOMETRY

Right rectangular prism X is similar to right rectangular prism Y. The surface area of right rectangular prism X is 58 square centimeters cm^2, and the surface area of right rectangular prism Y is 1,450 cm^2. The volume of right rectangular prism Y is 1,250 cubic centimeters cm^3. What is the sum of the volumes, in cm^3, of right rectangular prism X and right rectangular prism Y?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q3

GEOMETRY AND TRIGONOMETRY

A right circular cone has a volume of $71{,}148\pi$ cubic centimeters and the area of its base is $5{,}929\pi$ square centimeters. What is the slant height, in centimeters, of this cone?

A. 12

B. 36

C. 77

D. 85

Q4

GEOMETRY AND TRIGONOMETRY

A right triangle has sides of length $2\sqrt{2}$, $6\sqrt{2}$, and $\sqrt{80}$ units. What is the area of the triangle, in square units?

A. $8\sqrt{2} + \sqrt{80}$

B. 12

C. $24\sqrt{80}$

D. 24

Q5

GEOMETRY AND TRIGONOMETRY

A cube has a surface area of 54 square meters. What is the volume, in cubic meters, of the cube?

A. 18

B. 27

C. 36

D. 81

Q6

GRID-IN GEOMETRY AND TRIGONOMETRY

Triangles ABC and DEF are similar. Each side length of triangle ABC is 4 times the corresponding side length of triangle DEF. The area of triangle ABC is 270 square inches. What is the area, in square inches, of triangle DEF?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q7

GEOMETRY AND TRIGONOMETRY

Rectangle ABCD is similar to rectangle EFGH. The area of rectangle ABCD is 648 square inches, and the area of rectangle EFGH is 72 square inches. The length of the longest side of rectangle ABCD is 36 inches. What is the length, in inches, of the longest side of rectangle EFGH?

A. 4

B. 9

C. 12

D. 36

Q8

GRID-IN GEOMETRY AND TRIGONOMETRY

A rectangular poster has an area of 360 square inches. A copy of the poster is made in which the length and width of the original poster are each increased by 20%. What is the area of the copy, in square inches?

STUDENT-PRODUCED RESPONSE

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q9

GEOMETRY AND TRIGONOMETRY

A right rectangular prism has a base area of $24t$ square centimeters cm^2. The length of the base of the rectangular prism is $\frac{8}{3}$ cm, and the height of the rectangular prism is 15 cm. Which expression represents the surface area, in cm^2, of the right rectangular prism?

A. $48t + 160$

B. $318t + 80$

C. $1{,}968t + 80$

D. $360t$

Q10

GEOMETRY AND TRIGONOMETRY

Parallelogram $ABCD$ is similar to parallelogram $PQRS$. The length of each side of parallelogram $PQRS$ is 2 times the length of its corresponding side of parallelogram $ABCD$. The area of parallelogram $ABCD$ is 5 square centimeters. What is the area, in square centimeters, of parallelogram $PQRS$?

A. 7

B. 10

C. 20

D. 25